LN1 The Binomial Model

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Outlines

Textbook: Arbitrage Theory in Continuous Time: Chapter 2

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Model Description

Suppose we have:

- Two points in time: $t = 0$ and $t = 1$
- Two assets: a bond and a stock
  - B_t: the price of a bond at time t
    - ★ Bond price process B is given by

$$B_0 = 1$$
$$B_1 = 1 + R$$

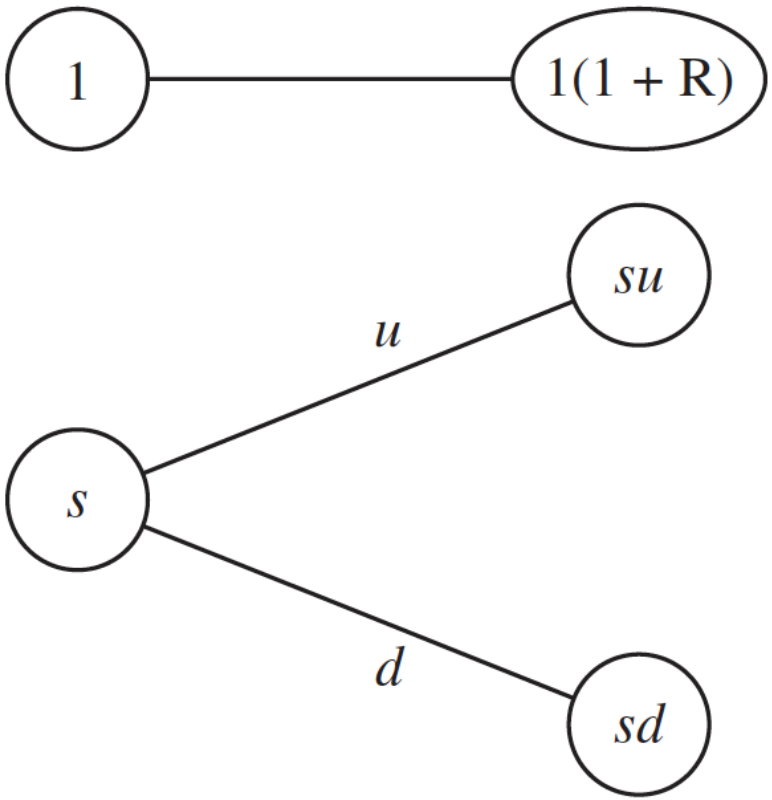
where R is the spot rate for the period.

  - S_t: the price of one share of the stock at time t
    - ★ Stock price process S is given by

$$S_0 = s,$$
$$S_1 = \begin{cases} s \cdot u, & \text{with probability } p_u \\ s \cdot d, & \text{with probability } p_d \end{cases}$$

Assume that s, u, d, p_u and p_d are known, and $d < u$, and $p_u + p_d = 1$.

Figure of One Period Price Tree



Model Description

It is often convenient to write the stock price process S as

$$S_0 = s,$$
$$S_1 = s \cdot Z$$

where Z is a stochastic variable defined as

$$Z = \begin{cases} u, \text{with probability } p_u \\ d, \text{with probability } p_d \end{cases}$$

Portfolios and Assumptions

Define a portfolio as a vector $h = (x, y)$, where x is the number of bonds we hold in the portfolio, and y is the number of units of stock held in the portfolio. We also have the following assumptions:

1. Short positions, as well as fractional holdings, are allowed. In mathematical terms this means that every $h \in R^2$ is an allowed portfolio.
2. There is no bid-ask spread, i.e. the selling price is equal to the buying price of all assets.
3. There are no transaction costs of trading.
4. The market is completely liquid, i.e. it is always possible to buy and/or sell unlimited quantities on the market. In particular it is possible to borrow unlimited amounts from the bank (by selling bonds short).

Value Process of Portfolio

Consider the market value of portfolio $h = (x, y)$ at $t = 0$ and $t = 1$.

Definition 1

The **value process** *of the portfolio h is defined by*

$$V_t^h = xB_t + yS_t, \text{ for } t = 0, 1$$

or, in more detail,

$$V_0^h = x + ys$$
$$V_1^h = x(1 + R) + ysZ$$

Arbitrage

Arbitrage portfolio is the central concepts of the theory.

Definition 2

The existence of **arbitrage** *portfolio is equivalent to a serious case of mispricing on the market.*

$$V_0^h = 0,$$
$$P(V_1^h \geq 0) = 1$$
$$P(V_1^h > 0) > 0$$

Conditions of Arbitrage Free Model

Proposition 1

The model above is free of arbitrage if and only if the following conditions hold:

$$d < (1 + R) < u. \tag{1}$$

Proof

$\Leftarrow$: Suppose $1 + R \geq u$. Then we have $s(1 + R) \geq su$, so it is always more profitable to invest in the bond than in the stock. Then we can construct a portfolio $h = (s, -1)$, i.e. we sell the stock and invest all the money in the bond. The portfolio has $V_0^h = 0$, and $V_1^h = s(1 + R) - sZ \geq 0$ by assumption. Therefore, this is an arbitrage portfolio.

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Proof of Proposition 1 Continued

$\Rightarrow$: Now assume that (1) holds. Consider an arbitrage portfolio $h = (x, y)$ at $t = 0$ such that $V_0^h = 0$. We thus have $x = -ys$. Then we can write the value of the portfolio at $t = 1$ as

$$V_1^h = \begin{cases} ys[u - (1 + R)], & \text{if Z = u} \\ ys[d - (1 + R)], & \text{if Z = d.} \end{cases}$$

If $y > 0$, h is an arbitrage strategy if and only if we have

$$u \geq 1 + R$$
$$d \geq 1 + R$$

but this is impossible because of condition (1).

Probability Measure Q

- Condition (1) implies that $1 + R$ is a convex combination of u and d, i.e.

$$1 + R = q_u \cdot u + q_d \cdot d$$

where $q_u, q_d \geq 0$, and $q_u + q_d = 1$.

- Weights q_u and q_d can be interpreted as probabilities for a new probability measure Q with the property $Q(Z = u) = q_u$, $Q(Z = d) = q_d$.
- Denote the expectation under measure Q as E^Q:

$$\frac{1}{1+R} E^Q[S_1] = \frac{1}{1+R}[q_u su + q_d sd] = \frac{1}{1+R} \cdot s(1+R) = s$$

- Therefore we have the following relation:

$$s = \frac{1}{1+R} E^Q[S_1]$$

Martingale Measure

$$s = \frac{1}{1+R} E^Q[S_1]$$

- This is in fact a **risk neutral** valuation formula, as it gives today's stock price as the discounted expected value of tomorrow's stock price.
- A probability measure with this property is called a **risk neutral measure**, or alternatively, a **risk adjusted measure**, or a **martingale measure**.

Definition 3

A probability measure Q is called a **martingale measure** *if the following condition holds:*

$$S_0 = \frac{1}{1+R} E^Q[S_1]$$

Condition of No Arbitrage in A New Way

Proposition 2

The market model is arbitrage free if and only if there exists a martingale measure Q.

Proposition 3

For the binomial model above, the martingale probabilities are given by

$$\begin{cases} q_u = \frac{(1+R)-d}{u-d} \\ q_d = \frac{u-(1+R)}{u-d} \end{cases}$$

Definition of A Contingent Claim

Assume that the market is arbitrage free.

Definition 4

A **contingent claim** *(financial derivative) is any stochastic variable X of the form $X = \Phi(Z)$, where Z is the stochastic variable driving the stock price process above.*

The function Φ is called the contract function.

Example of Contingent Claims

A typical example is European call option on the stock with strike price K.

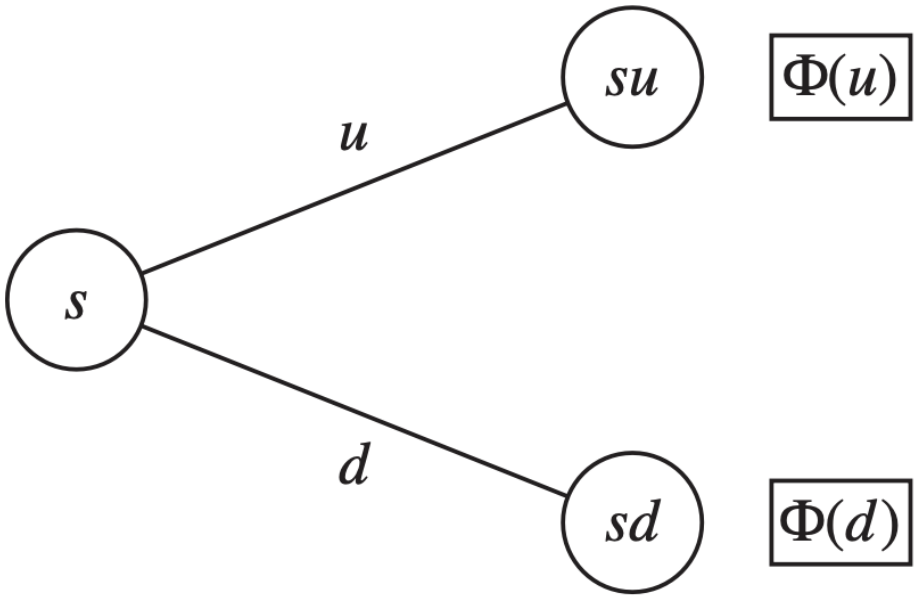
- Assume that $sd < K < su$.
  - If $S_1 > K$, we use the option and pay K to buy the stock and sell at su, then earn net profit of $su - K$.
  - If $S_1 < K$, we will not use the option.
- So X has the following dynamic:

$$X = \begin{cases} su - K, & \text{if } Z = u \\ 0, & \text{if } Z = d \end{cases}$$

and the contract function is

$$\Phi(u) = su - K,$$
$$\Phi(d) = 0$$

Figure of Example of Contingent Claims



Determine the "Fair" Price

Denote the price of X at time t by $\Pi(t; X)$. To avoid arbitrage we must have

$$\Pi(1; X) = X$$

To determine $\Pi(0; X)$, we need the following definition:

Definition 5

A given contingent claim X is said to be **reachable** *if there exists a portfolio h such that*

$$V_1^h = X$$

with probability 1. In that case we say that the portfolio h is a **hedging** *portfolio or a* **replicating** *portfolio. If all claims can be replicated we say that the market is* **complete**.

Pricing Principle

- From a financial point of view, there is no difference between holding X and portfolio h.
- Therefore, the price of X should equal the market value of the portfolio.
- Hence we have the following pricing principle:

Pricing principle 1

If a claim X is reachable with replicating portfolio h, then the only reasonable price process for X is given by

$$\Pi(t; X) = V_t^h, \ t = 0, 1.$$

Reasonable Price and Complete Market

What is a "reasonable price"?

Proposition 4

Suppose that a claim X is reachable with replicating portfolio h. Then any price at $t = 0$ of the claim X, other than V_0^h, will lead to an arbitrage possibility.

When is a market complete?

Proposition 5

Assume that the general binomial model is free of arbitrage. Then it is also complete.

Proof of Proposition 5

Proof Fix an arbitrary claim X with contract function Φ. We want to show that there exists a portfolio $h = (x, y)$ such that

$$V_1^h = \Phi(u), \text{ if } Z = u,$$
$$V_1^h = \Phi(d), \text{ if } Z = d$$

Specifically, the portfolio can be written as:

$$(1 + R)x + suy = \Phi(u).$$
$$(1 + R)x + sdy = \Phi(d).$$

The linear system has unique solution, by assumption $d < u$:

$$x = \frac{1}{1+R} \cdot \frac{u\Phi(d) - d\Phi(u)}{u - d} \tag{2}$$

$$y = \frac{1}{s} \cdot \frac{\Phi(u) - \Phi(d)}{u - d} \tag{3}$$

Risk Neutral Valuation

By pricing principle 1 and formulas (2) - (3), the price of at $t = 0$ is given by

$$\begin{aligned}\Pi(0; X) &= V_0^h \\ &= x + sy \\ &= \frac{1}{1+R}\left\{\frac{(1+R)-d}{u-d} \cdot \Phi(u) + \frac{u-(1+R)}{u-d} \cdot \Phi(d)\right\}\end{aligned}$$

Recall the martingale probabilities q_u and q_d defined in Proposition (3), so we can write the pricing formula above as

$$\Pi(0; x) = \frac{1}{1+R}\left\{\Phi(u) \cdot q_u + \Phi(d) \cdot q_d\right\}$$

Risk Neutral Valuation

Proposition 6

If the binomial model is free of arbitrage, then the arbitrage free price of a contingent claim X is given by

$$\Pi(0;X) = \frac{1}{1+R} E^Q[X] \tag{4}$$

Here the martingale measure Q is uniquely determined by the relation

$$S_0 = \frac{1}{1+R} E^Q[S_1] \tag{5}$$

and the explicit expression for q_u and q_d are given in Proposition 3.

Risk Neutral Valuation

Proposition 6 Continued

Furthermore the claim can be replicated using the portfolio

$$x = \frac{1}{1+R} \cdot \frac{u\Phi(d) - d\Phi(u)}{u-d} \tag{6}$$

$$y = \frac{1}{s} \cdot \frac{\Phi(u) - \Phi(d)}{u-d} \tag{7}$$

Example of One Period Model

We set $s = 100, u = 1.2, d = 0.8, p_u = 0.6, p_d = 0.4$ and, for computational simplicity, $R = 0$. By convention, the monetary unit is the US dollar.

- Thus we have the price dynamics

$$S_0 = 100,$$

$$S_1 = \begin{cases} 120, & \text{with probability } 0.6. \\ 80, & \text{with probability } 0.4. \end{cases}$$

Example of One Period Model

- We can compute the discounted expected value (under the objective probability measure P) of tomorrow's price

$$\frac{1}{1+R}E^P[S_1] = 1 \cdot [120 \cdot 0.6 + 80 \cdot 0.4] = 104.$$

- This is higher than the value of today's stock price of 100, so the market is risk averse.
- Since condition (1) obviously is satisfied we know that the market is arbitrage free.

Example of One Period Model

We consider a European call with strike price $K = 110$, so the claim X is given by

$$X = \begin{cases} 10, & \text{if } S_1 = 120. \\ 0, & \text{if } S_1 = 80. \end{cases}$$

- Using the naive method of computing the price as the discounted expected values under the objective probabilities, the price would be

$$\Pi(0; X) = \frac{1}{1+0}[10 \cdot 0.6 + 0 \cdot 0.4] = 6.$$

- Using the Proposition 3 it is easily seen that the martingale probabilities are given by $q_u = q_d = 0.5$, thus giving us the theoretical price

$$\Pi(0; X) = \frac{1}{1+0}[10 \cdot 0.5 + 0 \cdot 0.5] = 5.$$

Example of One Period Model

If our theory is correct, we should be able to replicate the option. From the proposition 6, the replicating portfolio is given by

$$x = \frac{1.2 \cdot 0 - 0.8 \cdot 10}{1.2 - 0.8} = -20,$$
$$y = \frac{1}{100} \cdot \frac{10 - 0}{1.2 - 0.8} = \frac{1}{4}.$$

- In everyday terms this means that the replicating portfolio is formed by borrowing \$20 from the bank, and investing this money in a quarter of a share in the stock.
  - The net value of the portfolio at $t = 0$ is \$5.
  - At $t = 1$ the value is given by

$$V_1^h = -20 + \frac{1}{4} \cdot 120 = 10, \text{if } S_1 = 120,$$
$$V_1^h = -20 + \frac{1}{4} \cdot 80 =, \text{if } S_1 = 80.$$

Example of One Period Model

- If anyone buys the option from us for the price \$6, then we can make a riskless profit (an arbitrage profit) by
  - selling the option to obtain six dollars,
  - investing five in the replicating portfolio,
  - investing the remaining one in the bank.

  At time $t = 1$, the claims of the buyer of the option are completely balanced by the value of the replicating portfolio, and we still have one dollar ivested in the bank.
- If someone is willing to sell the option to us at a price lower than five dollars, we can also make an arbitrage profit by selling the portfolio short.

Model Description

The multiperiod model is a discrete time model with the time index t running from $t = 0$ to $t = T$. As before there are two underlying assets: bond and stock. The bond price dynamics B_t is given by

$$B_{n+1} = (1 + R)B_n$$
$$B_0 = 1$$

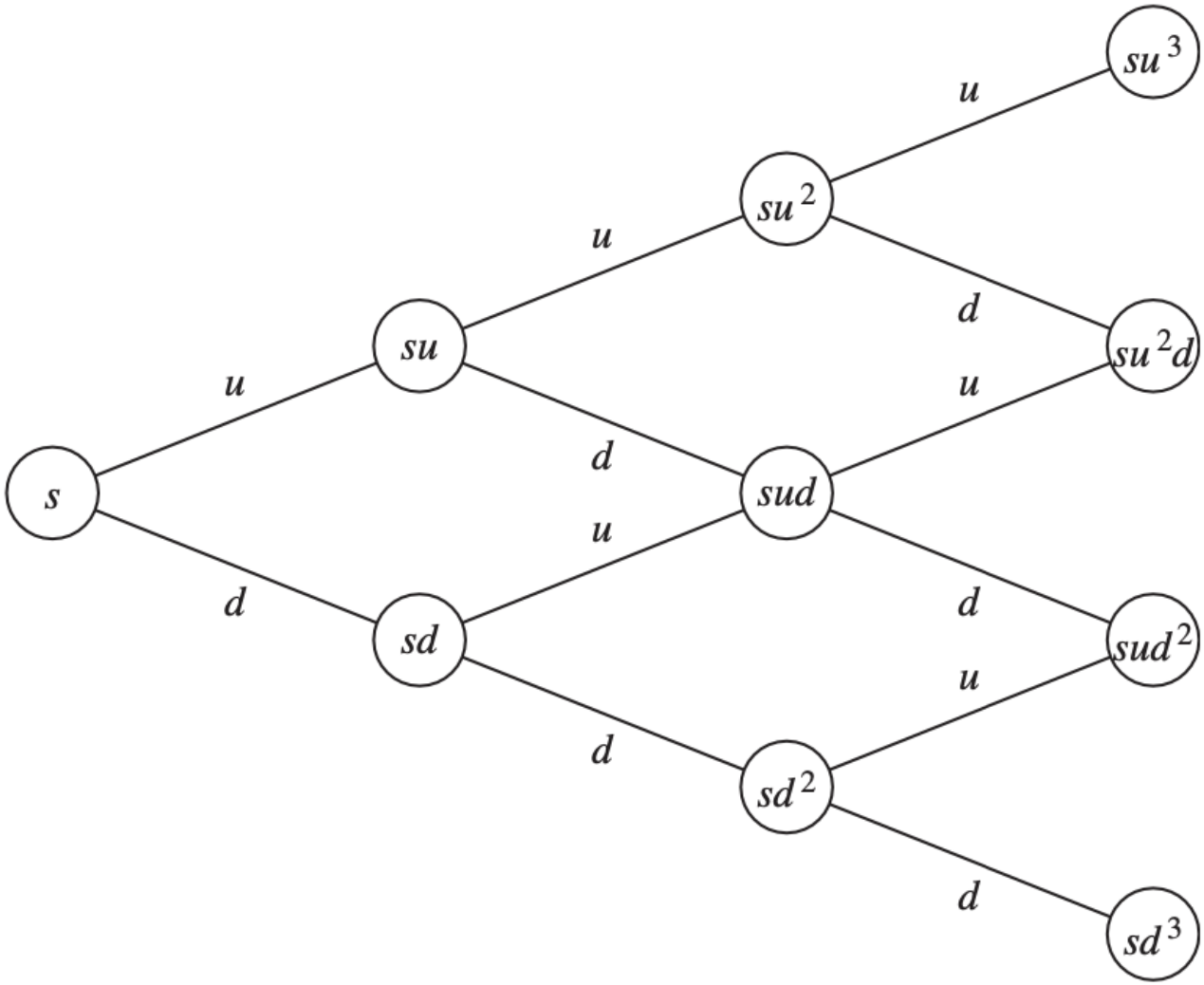
The stock price dynamics S_t is given by

$$S_{n+1} = S_n \cdot Z_n$$
$$S_0 = s$$

here $Z_0, \cdots, Z_{T-1}$ are assumed to be i.i.d (independent and identically distributed) random variables, taking two values u and d with probabilities

$$P(Z_n = u) = p_u$$
$$P(Z_n = d) = p_d$$

Figure of Price Dynamics



Define A Dynamic Portfolio Strategy

Now we define the dynamic portfolio strategy:

Definition 6

A **portfolio strategy** *is a stochastic process*

$$\{h_t = (x_t, y_t); t = 1, \cdot, T\}$$

such that h_t *is a function of* $S_0, S_1, \cdots, S_{t-1}$. *For a given portfolio strategy* h *we set* $h_0 = h_1$ *by convention. The* **value process** *corresponding to the portfolio* h *is defined by*

$$V_t^h = x_t(1 + R) + y_t S_t$$

Interpretation of Definition 6

- x_t is the amount of money invested in the bank at time $t-1$ and kept until time t
- y_t is the number of shares bought at time $t-1$ and kept until time t.
- V_t^h is the market value of portfolio (x_t, y_t) (which has been held since $t-1$) at time t.
- We allow the portfolio strategy to be a contingent strategy, i.e. the portfolio we buy at t is allowed to depend on all information we have collected by observing the evolution of the stock price up to time t. We are, however, not allowed to look into the future.

Self Financing

The portfolios that are of primary interest is self-financing portfolios.

In practical terms in a self-financing portfolio strategy the accession of a new asset has to be financed through the sale of some other asset. The mathematical definition is as follows.

Definition 7

A portfolio strategy h is said to be **self financing** *if the following condition holds for all $t = 0, \cdots, T-1$*

$$x_t(1+R) + y_t S_t = x_{t+1} + y_{t+1} S_t$$

The interpretation of this definition is that at time t, the market value of the "old" portfolio (x_t, y_t) equals the purchase value of the new portfolio (x_{t+1}, y_{t+1}).

Arbitrage Possibility

Multi-period version of an arbitrage possibility is defined by

Definition 8

An **arbitrage** *possibility is a self-financing portfolio h with the properties*

$$V_0^h = 0$$
$$P(V_T^h \geq 0) = 1$$
$$P(V_T^h > 0) > 0$$

Necessary Condition for Absence of Arbitrage

Lemma 1

If the model is free of arbitrage then the following conditions necessarily must hold

$$d \leq (1 + R) \leq u \tag{8}$$

The condition is in fact also sufficient for absence of arbitrage. We will prove it later. From now on we assume that $d < u$, and that the condition (8) holds.

Martingale Probabilities

Definition 9

The martingale probabilities q_u and q_d are defined as the probabilities for which the relation

$$s = \frac{1}{1+R} E^Q[S_{t+1}|S_t = s]$$

holds

Proposition 7

The martingale probabilities are given by

$$\begin{cases} q_u = \frac{(1+R)-d}{u-d} \\ q_d = \frac{u-(1+R)}{u-d} \end{cases}$$

Contingent Claims

Definition 10

A **contingent claim** *is a stochastic variable* X *of the form*

$$X = \Phi(S_T)$$

where the **contract function** Φ *is some given real valued function.*

One main problem is how to find a reasonable price process

$$\{\Pi(t; X); t = 0, \cdots, T\}$$

for a given claim X.

Replicating Portfolios

Definition 11

A given contingent claim X is said to be **reachable** *if there exists a self-financing portfolio h such that*

$$V_T^h = X$$

with probability 1. In that case we say that the portfolio h is a **hedging** *portfolio or a* **replicating** *portfolio. If all claims can be replicated we say that the market is (dynamically)* **complete**.

Pricing Principle

The pricing principle for reachable claims is:

Pricing principle 2

If a claim X is reachable with replicating (self-financing) portfolio h, then the only reasonable price process for X is given by

$$\Pi(t; X) = V_t^h, \quad t = 0, 1, \cdots T$$

This pricing principle leads to the following proposition:

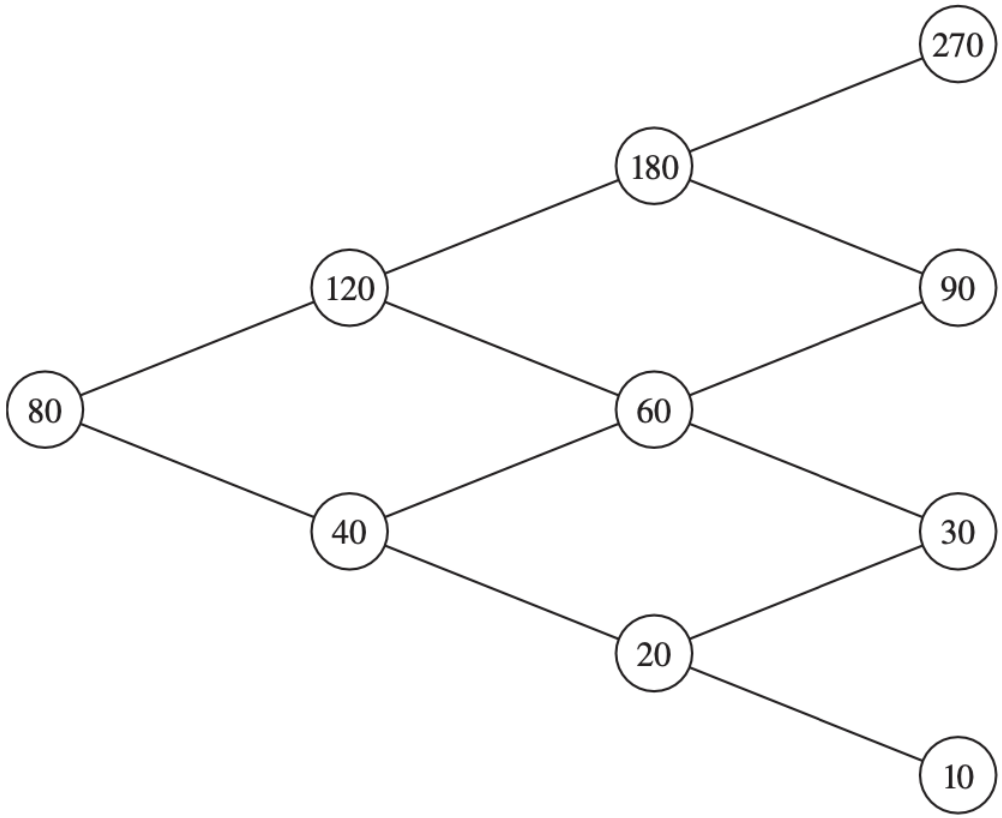
Proposition 8

Suppose that X is reachable using the portfolio h. Suppose furthermore that, at some time t, it is possible to buy X at a price cheaper than (or to sell it at a price higher than) V_t^h. Then it is possible to make an arbitrage profit.

Example of A Multiperiod Binomial Model

We set $T = 3, S_0 = 80, u = 1.5, d = 0.5, p_u = 0.6, p_d = 0.4, R = 0$.

- The dynamics of the stock price can be illustrated using the binomial tree.



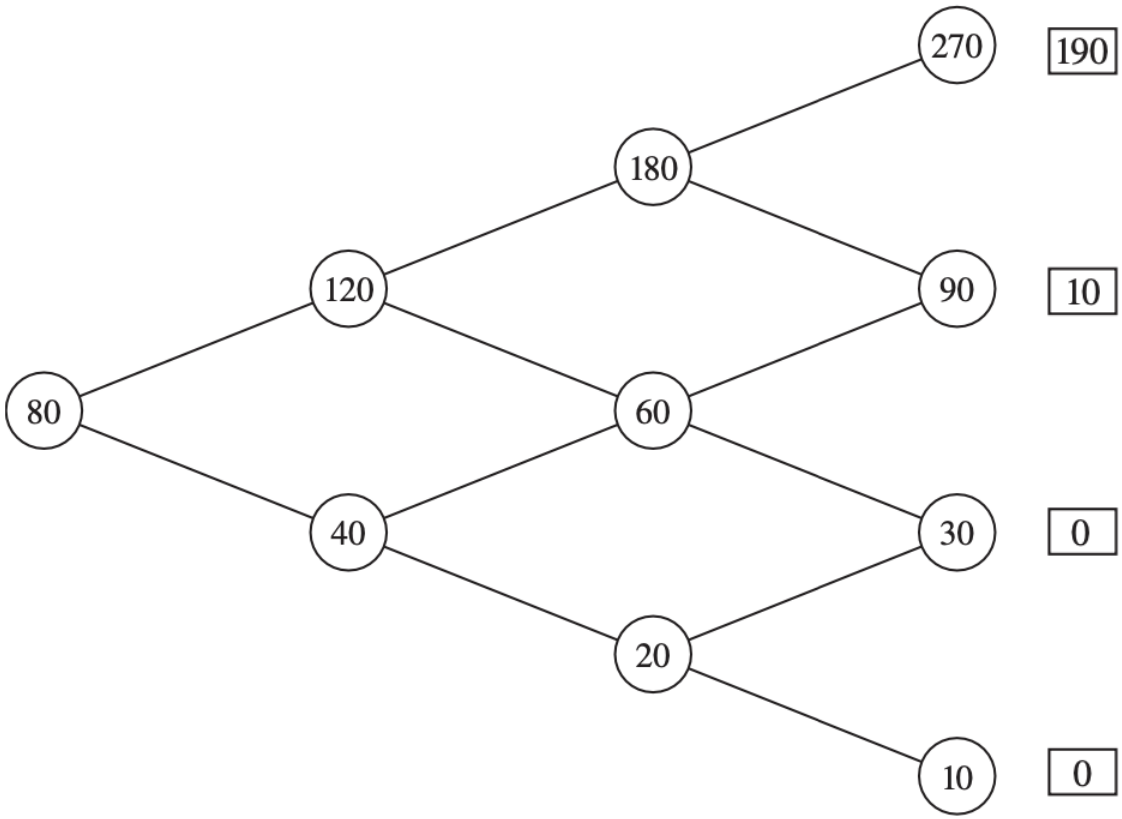
Example of A Multiperiod Binomial Model

We now consider a European call on the underlying stock. The date of expiration of the option is $T = 3$, and the strike price is chosen to be $K = 80$. Formally this claim can be described as $X = max[S_T - K, 0]$.

- In the following, we will show that this particular claim can be replicated. The idea is to use induction on the time variable and to work backwards in the tree from the leaves at $t = T$ to the root at $t = 0$.

Example of A Multiperiod Binomial Model

- We start by computing the price of the option at the date of expiration using the relation $\Pi(T; X) = X$. The boxed numbers indicate the price of the claim.



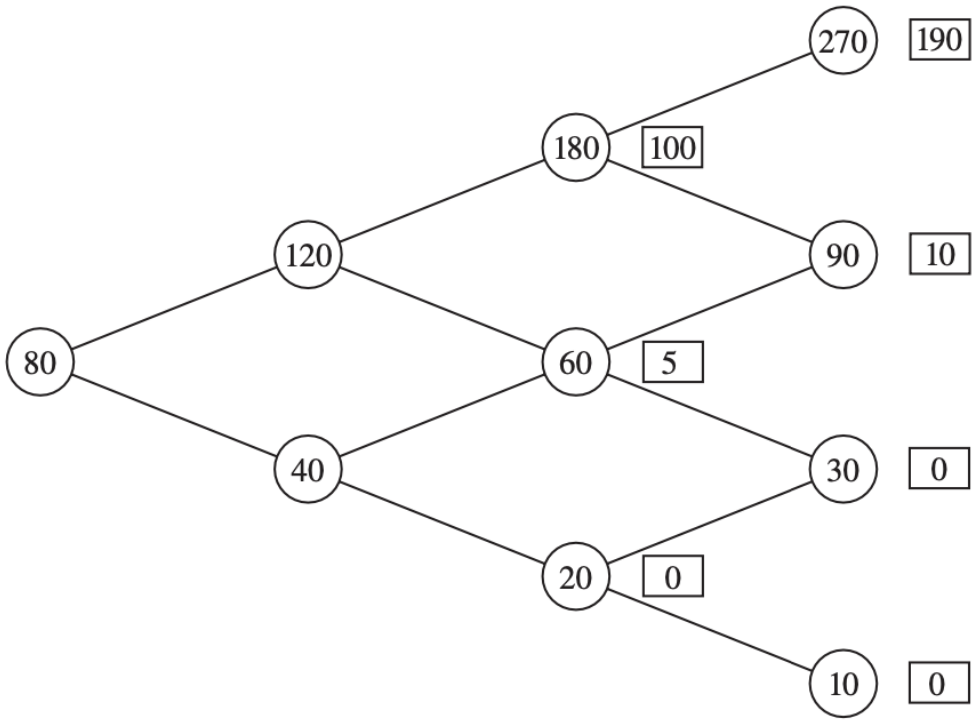
Example of A Multiperiod Binomial Model

- Our problem is thus that of replicating the boxed payoff structure at $t = 3$.
- If we look closer at some node at $t = 2$, e.g. at the node $S_2 = 180$, we see a simple one period binomial model, and it follows directly from the one period theory that the payoff structure can indeed be replicated from the node $S_2 = 180$.
- We can in fact compute the cost of this replicating portfolio by risk neutral valuation, and since the martingale probabilities for this example are given by $q_u = q_d = 0.5$ the cost of the replicating portfolio is

$$\frac{1}{1+0}[190 \cdot 0.5 + 10 \cdot 0.5] = 100.$$

Example of A Multiperiod Binomial Model

- In the same way we can consider all the other nodes at $t = 2$, and compute the cost of the corresponding replicating portfolios. The result is the set of boxed numbers at $t = 2$.



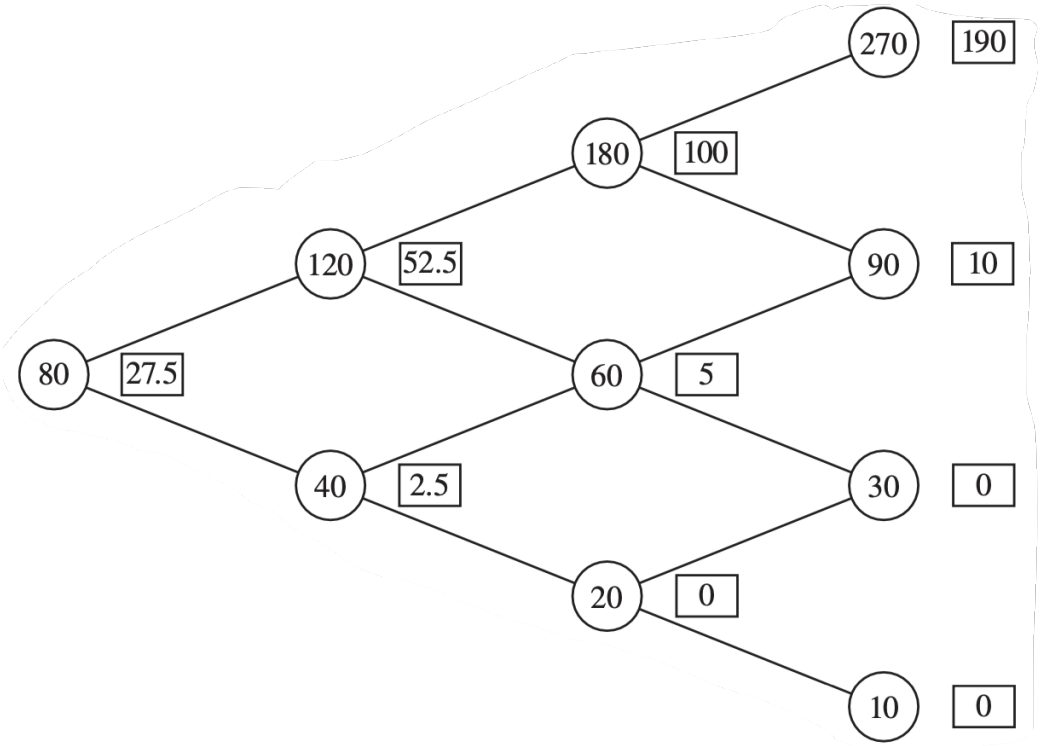
Example of A Multiperiod Binomial Model

- What we have done by this procedure is to show that if we can find a self-financing portfolio which replicates the boxed payoff structure at $t = 2$, then it is in fact possible to replicate the original claim at $t = 3$.
- We have thus reduced the problem in the time variable, and from now on we simply reproduce the construction above, but at t $= 1$.
- Take, for example, the node $S_1 = 40$. From the point of view of this node we have a one period model, and by risk neutral valuation we can replicate the payoff structure using a portfolio, which at the node $S_1 = 40$ will cost

$$\frac{1}{1+0}[5 \cdot 0.5 + 0 \cdot 0.5] = 2.5$$

Example of A Multiperiod Binomial Model

- In this manner we fill the nodes at $t = 1$ with boxed portfolio costs, and then we carry out the same construction again at $t = 0$.



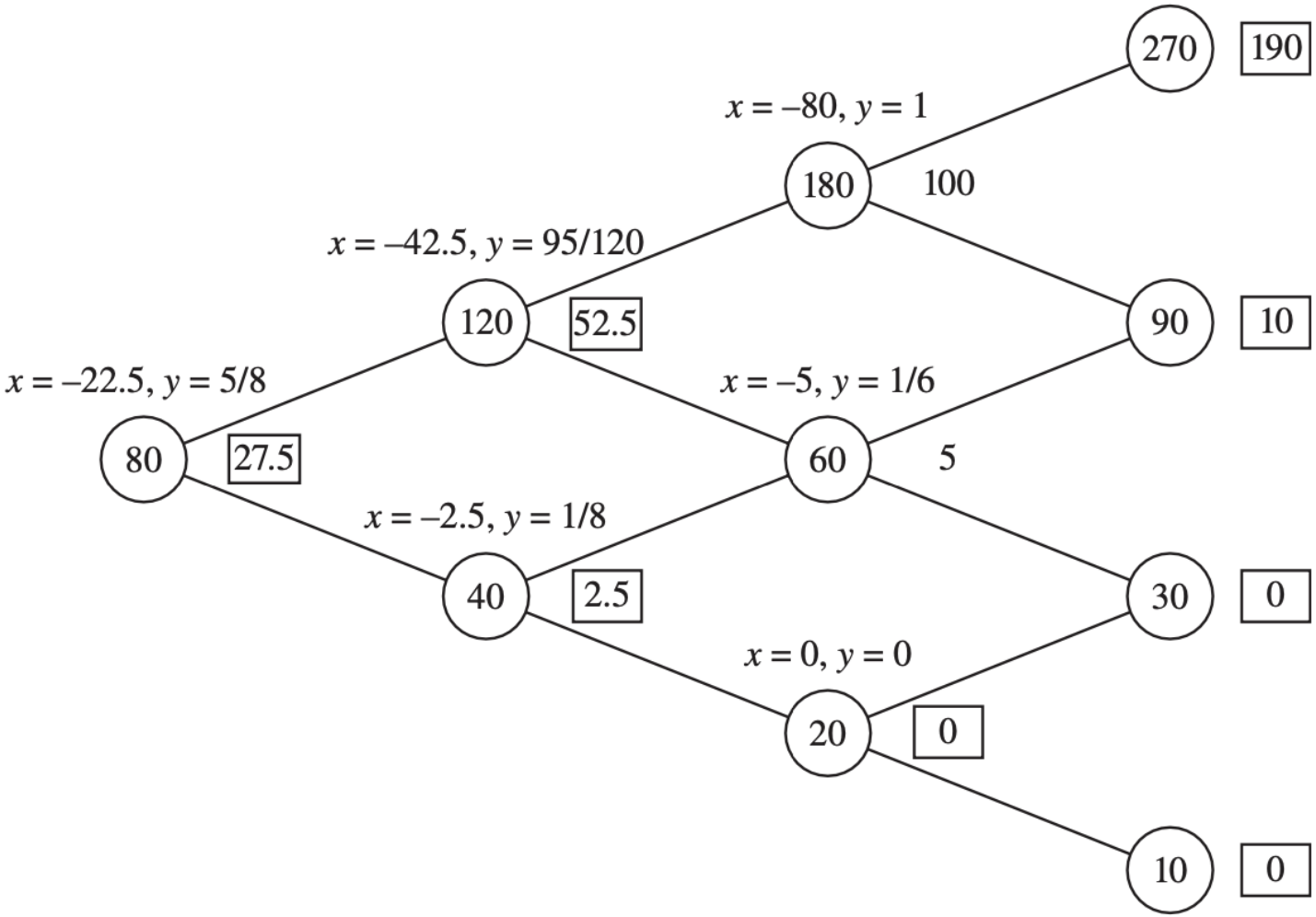
- We have thus proved that it is in fact possible to replicate the European call option at an initial cost of 27.5.

Example of A Multiperiod Binomial Model

- We can use Proposition 6 to compute the hedging portfolio as $x_1 = -22.5, y_1 = 5/8$ at $t = 0$ to reproduce the boxed claim $(52.5, 2.5)$ at $t = 1$. The cost of this portfolio is exactly 27.5.
- Suppose that the price now moves to $S_1 = 120$. Then our portfolio is worth
$$-22.5 \cdot (1 + 0) + \frac{5}{8} \cdot 120 = 52.5.$$
- Since we now are facing the claim $(100, 5)$ at $t = 2$ we can again use Proposition 6 to calculate the hedging portfolio as $x_1 = -42.5, y_1 = 95/120$, and the cost of this portfolio equals the value of our old portfolio, i.e. 52.5. Thus it is really possible to rebalance the portfolio in a self-financing manner.
- The process can continue...

Example of A Multiperiod Binomial Model

In the figure below, we have computed the hedging portfolio at each node.



Binomial Algorithm – Notation

The value of the price process at time t can be written as

$$S_t = su^k d^{t-k}, k = 0, ..., t$$

where k denotes the number of up-moves that have occurred. Thus each node in the binomial tree can be represented by a pair (t, k) with $k = 0, ..., t$.

Binomial Algorithm

Proposition 9

(Binomial algorithm) *Consider a T-claim* $X = \Phi(S_T)$. *Then this claim can be replicated using a self-financing portfolio. If* $V_t(k)$ *denotes the value of the portfolio at the node* (t, k) *then* $V_t(k)$ *can be computed recursively by the scheme*

$$\begin{cases} V_t(k) = \frac{1}{1+R}\{q_u V_{t+1}(k+1) + q_d V_{t+1}(k)\} \\ V_T(k) = \Phi(s u^k d^{T-k}) \end{cases}$$

where the martingale probabilities q_u *and* q_d *are given by*

$$\begin{cases} q_u = \frac{(1+R)-d}{u-d} \\ q_d = \frac{u-(1+R)}{u-d} \end{cases}$$

Binomial Algorithm Continued

Proposition 9 Continued

With the notation as above, the hedging portfolio is given by

$$\begin{cases} x_t(k) = \frac{1}{1+R} \cdot \frac{uV_t(k) - dV_t(k+1)}{u-d} \\ y_t(k) = \frac{1}{S_{t-1}} \cdot \frac{V_t(k+1) - V_t(k)}{u-d} \end{cases}$$

In particular, the arbitrage free price of the claim at $t = 0$ is given by $V_0(0)$.

Risk Neutral Valuation Formula

Hence we obtain a risk neutral valuation formula

Proposition 10

The arbitrage free price at $t = 0$ of a T-claim X is given by

$$\Pi(0; X) = \frac{1}{(1+R)^T} \cdot E^Q[X]$$

where Q denotes the martingale measure, or more explicitly

$$\Pi(0; X) = \frac{1}{(1+R)^T} \cdot \sum_{k=0}^{T} \binom{T}{k} q_u^k q_d^{T-k} \Phi(s u^k d^{T-k})$$

Proof of Proposition 10

Proof The first formula follows directly from the algorithm above. Let Y denote the number of up-moves in the tree, then

$$X = \Phi(S_T) = \Phi(su^Y d^{T-Y})$$

Then use the fact that Y has binomial distribution to get the second formula. □